Joint Probability Distribution

- The function $f(x,y)$ is a joint probability distribution or probability mass function of the discrete random variables X & Y if

1. $f(x,y) \geq 0, \forall x,y$
2. $\sum_{x} \sum_{y} f(x,y) = 1$
3. $P(X = x, Y = y) = f(x,y)$

- For any region A in the xy plane,
$P[(X,Y) \in A] = \sum \sum_{A} f(x,y)$

- The funcn $f(x,y)$ is a joint density funcn of the continuous random variable X & Y if –

1. $f(x,y) \geq 0 \;\forall x,y$
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dx\, dy = 1$
3. $P((X,Y) \in A) = \int \int_{A} f(x,y)\, dx\, dy$

for any region A in the xy plane

- The marginal distributions of X alone & of Y alone are —

$$g(x) = \sum_y f(x,y)$$
$$h(y) = \sum_x f(x,y)$$

$\Bigg\}$ for the discrete case

$$g(x) = \int_{-\infty}^{\infty} f(x,y)\, dy$$
$$h(y) = \int_{-\infty}^{\infty} f(x,y)\, dx$$

$\Bigg\}$ for the continuous case

- $P(a<X<b) = \int_a^b g(x)\, dx$

- Let X & Y be two random variables, discrete or continuous. The conditional distribuⁿ of the random variable Y given that $X = x$ is,

$$f(y|x) = \frac{f(x,y)}{g(x)}, \quad g(x) > 0$$

Similarly, the conditional distribuⁿ of X given that $Y = y$ is

$$f(x|y) = \frac{f(x,y)}{h(y)}, \quad h(y) > 0$$

- $P(a < X < b \mid Y = y) = \int_a^b f(x|y)\, dx$

- Let X & Y be two random variables, discrete or continuous, with joint probability distribuⁿ $f(x,y)$ & marginal distributions $g(x)$ & $h(y)$, respectively. The random variables X & Y are said to be statistically independent, iff.

$$\boxed{f(x,y) = g(x)\, h(y)}$$

$\forall x, y$ within their range.